

Week 8, Day 1 — presenter notes

Slides: [week-08-day-01.html](#) → F · arrows · N · P.

Pointers, memory pictures, repository layer. Nil crash → fix → move map into `StudentRepository`. No JWT / Postgres.

Timing (3–3.5 hours)

Clock	Block	Slides
0:00–0:15	Week 7 create/list check	1–2
0:15–0:30	Outcomes + SE framing	3–4
0:30–1:20	Pointers, nil, receivers, value/ref, stack/heap/GC	5–12
1:20–2:20	Nil crash live + extract repo	13–18
2:20–end	Lab extract	19–end

Cut escape-analysis `-m` and make/new if create/list extract eats time. Never skip the nil panic.

Live demo

1. Register routes with a nil `*StudentHandler` → panic on first request
2. Fix with `NewStudentHandler(repo)`
3. Move map into `internal/repository/student_repository.go`
4. Handler holds `*StudentRepository`; curl create/list still green

Lab gate

- [] Nil crash seen and fixed
- [] Repo owns the map
- [] Handler has no map fields
- [] Postman create + list + get
- [] Health still green

Homework

PR with handler → repository. Memory diagram in the PR (notebook photo OK).

What not to teach

Postgres/ `pgx`, JWT, deep escape analysis, repository interfaces (concrete type is fine).